



Fazley Rabbi

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ABOUT ME

I am a passionate Computer Science graduate and IT professional who thrives on solving real world software engineering challenges and exploring data driven research. Despite facing significant personal and external disruptions during my undergraduate years including severe family bereavement and pandemic related challenges I remained deeply committed to my growth. This resilience pushed me to co-author two accepted international research papers in Deep Learning and Computer Vision. Alongside my research focus, I have accumulated valuable hands on industry experience building modern, type-safe web applications using Next.js and TypeScript. I am now eager to bring this unique blend of technical skills, research insights, and strong determination to an impactful graduate program or global technical role.

WORK EXPERIENCE

SOFTWARE ENGINEER (FRONTEND INTERN) – MEDISTACK – 01/01/2026 – 30/04/2026 – DHAKA, BANGLADESH

Website: <https://medistack.net/>

- 1. Frontend Web Development: Actively involved in developing responsive, high-performance single-page web applications utilizing ReactJS, Next.js, and TypeScript.
- 2. UI/UX Implementation: Engineered clean, intuitive user interfaces by leveraging Tailwind CSS, Vanilla CSS, and modern UI component libraries like Shadcn UI.
- 3. API Integration: Seamlessly connected frontend consumer modules with production-level backend services via REST API integrations and state management pipelines.
- 4. Agile Collaboration: Collaborated closely in a fast-paced environment with cross-functional software teams to debug interfaces, translate feature criteria into code, and deliver scalable frontend architectures with high dedication and professionalism.

EDUCATION AND TRAINING

01/01/2020 – 31/12/2025 Dhaka, Bangladesh

BSC IN COMPUTER SCIENCE & ENGINEERING Daffodil International University

- 1. Undergraduate Thesis & Core Research:** Co-authored a medical AI research study titled "Comparative Analysis of Boosting Ensembles and Neural Networks with Custom Models for Chronic Kidney Disease Prediction". Evaluated baseline machine learning algorithms (AdaBoost, Gradient Boosting, KNN, MLP) against custom architectures using a clinical dataset of 2,500 patient records.
 - 2. Core Contributions:** Developed and empirically tuned a novel Custom Weighted Ensemble Model (combining Optimized Decision Tree and KNN). This framework successfully neutralized noisy clinical data, achieved a superior diagnostic accuracy and recall of 79.00%, and minimized false negatives for early clinical diagnosis.
 - 3. System Deployment:** Successfully translated the machine learning pipeline into a production-ready, cloud-based web application for real-time kidney health monitoring and interactive clinical support.
- Core Computer Science Competencies: Advanced Data Structures, Algorithms, Neural Networks, Computer Vision, Deep Learning, Database Management Systems (DBMS), and Software Engineering.

Field of study Computer Science | **Final grade** 2.78 / 4.00 | **Level in EQF** EQF level 6 |

Thesis Comparative Analysis of Boosting Ensembles and Neural Networks with Custom Models for Chronic Kidney Disease Prediction

Field of study Science | **Final grade** 3.58/5.00 | **Level in EQF** EQF level 4

Field of study Science | **Final grade** 5.00/5.00 | **Level in EQF** EQF level 3

● PUBLICATIONS

2026

A Hybrid DenseNet-Vision Transformer Framework for Diabetic Retinopathy Classification

- 1.Co-authored (4th Author) a medical Artificial Intelligence (AI) and computer vision research study focused on automated ophthalmic screening.
- 2.Contributed to the development and evaluation of a cutting-edge hybrid deep learning model combining Convolutional Neural Networks (DenseNet) and Vision Transformers (ViTs) to accurately classify multi-stage Diabetic Retinopathy from retinal fundus images.
- 3.Assisted the research team in pipeline architecture verification, hyperparameter tuning, and cross-validation performance analytics against baseline medical imaging models.

Authors: Fazley Rabbi , et al. | **Journal Name:** IEEE International Conference | **Publisher:** IEEE

● PROJECTS

01/04/2025 – 31/12/2025

Chronic Kidney Disease Predictor

Designed, developed, and deployed a fully functional production-ready web application utilizing a custom predictive Machine Learning model for early-stage kidney disease risk screening. Engineered a highly responsive and modern user interface using Next.js, ReactJS, and TypeScript, optimized with Tailwind CSS and styled components for a seamless user experience. Integrated the frontend system with the underlying Python-based ML pipeline via efficient REST API architectures, enabling the system to accept 25 clinical parameters and visualize complex diagnostic matrices in real-time. Focused heavily on minimizing false negatives within the user workflow to maximize patient safety and clinical decision support efficiency.

Link <https://ckd-predictor-25.netlify.app/>

01/01/2026 – 03/03/2026

Mobile Shop – E-Commerce Web Application

Driven by a deep passion for frontend architecture, I independently built this modern mobile e-commerce web application to master production-grade web technologies. To challenge myself with real-world data flows, I integrated an external REST API to dynamically fetch smartphone catalogs and implemented advanced frontend logic like debounced search, brand-based filtering, and a consistent pricing algorithm. I focused heavily on delivering a premium user experience by crafting a polished, Apple-inspired layout with Tailwind CSS v4, complete with smooth skeleton loaders, dynamic variant selectors (color and storage), and a fully interactive 3D phone model viewer via Google Model Viewer. To simulate a complete e-commerce lifecycle without a backend, I engineered a dynamic shopping cart, a user wishlist system, and a multi-step checkout workflow entirely persisted through browser localStorage with seamless cross-tab synchronization. This project significantly matured my clean component styling, client-side state management, and responsive UI engineering.

Link <https://mobile-shop-client-hazel.vercel.app/>

01/03/2026 – 30/04/2026

Food Delivery Web Application

Driven by a strong interest in engineering seamless user experiences, I built this responsive client-side food delivery frontend application to master production-grade component structuring and modern data flows. Developed using Next.js 16 (App Router), React 19, and TypeScript, the application consumes TheMealDB REST API to deliver a dynamic, Foodpanda-style ordering lifecycle. I engineered complex frontend architectures including parallel data fetching with Promise.all for optimized initial loading, loading skeleton screens to improve perceived performance, and a custom debounced search algorithm for instantaneous menu discovery. To simulate a complete transactional flow without a

backend database, I built an intricate client-side checkout wizard, responsive dynamic recipe routing, and managed data persistence for user wishlists and the shopping cart via browser localStorage with real-time cross-tab synchronization. Styling was built with Tailwind CSS v4 to guarantee high-fidelity responsive layouts across all viewports.

Link restraurent-client.vercel.app

● **LANGUAGE SKILLS**

Mother tongue(s): **BENGALI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B1	B1	B1	B1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **DIGITAL SKILLS**

Frontend Development & Machine Learning

- 1.Languages & Frameworks: Next.js, ReactJS, TypeScript, JavaScript, HTML5, CSS3, Python.
- 2.Libraries & UI Tools: Tailwind CSS, Shadcn UI, REST API Integration.
- 3.Tools & Platforms: Git, GitHub, VS Code, Netlify.